

PROJECT: AWS COST MONITORING & BUDGET ALERTS

Services: AWS Budgets, Cost Explorer, Lambda, SES

Goal: Automatically notify when budget thresholds are met.

Skills: Billing alerts, email notifications, tagging strategies.

Definition: AWS Cost Monitoring & Budget Alerts is a cloud cost management system that monitors AWS resource usage and tracks overall spending. It allows users to define a budget limit and automatically receive alerts when the spending reaches specific thresholds. This helps users stay aware of their cloud expenses and avoid unexpected billing.

Scope of the Project:

- Implement cost tracking using AWS Budgets and Cost Explorer to monitor AWS usage and spending.
- Configure real-time alerts via Amazon CloudWatch and Amazon SNS for threshold-based notifications.
- Automate advanced notifications using AWS Lambda and Amazon SES for customized alerting.
- Apply tagging strategies to track and optimize costs across projects, environments, and resources efficiently.

Methodologies:

- Enabled billing insights using Cost Explorer and configured budgets in AWS Budgets to define spending limits.
- Created threshold-based alarms in Amazon CloudWatch to continuously monitor cost metrics.
- Integrated notifications via Amazon SNS and automated alerts using AWS Lambda with Amazon SES.
- Implemented resource tagging and cost allocation strategies to analyze and optimize expenses across different services and environments.

Services Used in this Project:

1. AWS Budgets – To set spending limits and receive budget alerts.
2. Cost Explorer – To analyze and visualize AWS usage and cost trends.
3. Amazon CloudWatch – To create alarms based on billing metrics.
4. Amazon SNS – To send alert notifications via email/SMS.
5. AWS Lambda – To automate alert handling and custom actions.
6. Amazon SES – To send customized email alerts through Lambda.

Architecture Diagram:



Implementation and Execution of AWS Cost Monitoring & Budget Alerts.

STEP 1: Login to AWS Console

- Region doesn't matter (Billing is global)

STEP 2: Enable Cost Explorer

- In Billing Dashboard → Cost Explorer
- Click Enable Cost Explorer
- Wait 24 hours (data will populate gradually)

Billing and Cost Management > Cost Explorer > New cost and usage report

New cost and usage report

Explore your costs with AI New

What was my untagged resources cost for last month? Show my daily cost and usage for compute hours What Ask question

Cost and usage overview Info

Total cost	Average monthly cost	Service count
-\$0.00	-\$0.00	20

Cost and usage graph

Costs (\$)

10

5

Report parameters

Time

Standard Compare

Date Range

2025-10-01 — 2026-03-31

Displaying last 6 months

Granularity

Monthly

Group by

Dimension

Service

Filters Info

Applied filters (0) Clear all

Service

Choose services

Linked account

Choose linked accounts

Region

Choose region

STEP 3: Create AWS Budget (Cost Threshold)

- Open Billing Dashboard
- Click Budgets
- Click Create budget

Budget Setup:

- Budget type: Cost budget
- Period: Monthly

Budget amount:

Example: \$1

- Click Next

Billing and Cost Management > Budgets > Budget details

MyMonthlyBudget Info

Budget health Info

Current vs. budgeted

Amount spent: \$22.76 of \$100.00 22.75%

Forecasted vs budgeted (MTD)

Amount spent: \$77.80 of \$100.00 77.80%

Alerts Info

Thresholds

OK

Actions

View all alerts

Details

Health status	Budget amount	Start date
Healthy	\$100.00	2026-04-01
Budget type	Period	End date
Cost budget	Monthly	-

Additional budget parameters

Budget history Alerts Tags Billing View

STEP 4: Alerts Setup Create alert:

- Alert 1 (Warning)

- Threshold: 85%

- Trigger: Actual cost Email recipients:

- your email (for now)

Click Create Budget.

Billing and Cost Management > Budgets > Budget details > Alert details

Your budget [MyMonthlyBudget](#) has been updated successfully.

Alert details - Actual cost > 85% [Info](#)

[Delete alert](#) [Edit alert](#)

Alert definition

When your actual cost is greater than 85% (\$85.00) of your budgeted amount (\$100.00), the alert threshold will be exceeded.

Status

Threshold	Actions
✔ Not exceeded	-

Alert settings

Budget name attached to MyMonthlyBudget	Threshold Greater than 85% of budgeted amount	Email recipients 1 email recipient
Budget type Cost	Threshold measured against Actual cost	Amazon SNS Not enabled

Billing and Cost Management > Budgets > Overview

Your budget [MyMonthlyBudget](#) has been updated successfully.

Budgets (1/1) [Info](#)

[Download CSV](#) [Actions](#) [Create budget](#)

Find a budget Type - Show all budgets

☒	Name	Thresholds	Health status	Billing View	Budget	Amount used	Forecasted ...	Current vs. buc
☒	MyMonthlyBudget	✔ OK	✔ Healthy	Primary View	\$100.00	\$23.94	\$70.02	

Billing and Cost Management > Budgets > Edit budget

Your budget [MyMonthlyBudget](#) has been updated successfully.

Set your budget [Info](#)

Step 1: Set your budget (selected)
Step 2: Configure alerts
Step 3 - Optional: Attach actions
Step 4: Review

How to set up your budget

Step 1: Enter your budget details

Define the budget name.

Step 2: Set budget amount

Select the period and whether you would like to have a fixed budget or to specify a budget plan, then enter your budget amount.

Step 3: Scope your budget - optional

Add dimensions of data to narrow on a set of cost information. For example, you could select a number of AWS services to track as part of this budget.

Budget preview

Cost Data
Apr 2025 - Apr 2026 (MTD) | Unblended costs

Actual cost Budget Alert #1 (actual) Alert #2 (forecasted) Alert #3 (actual)

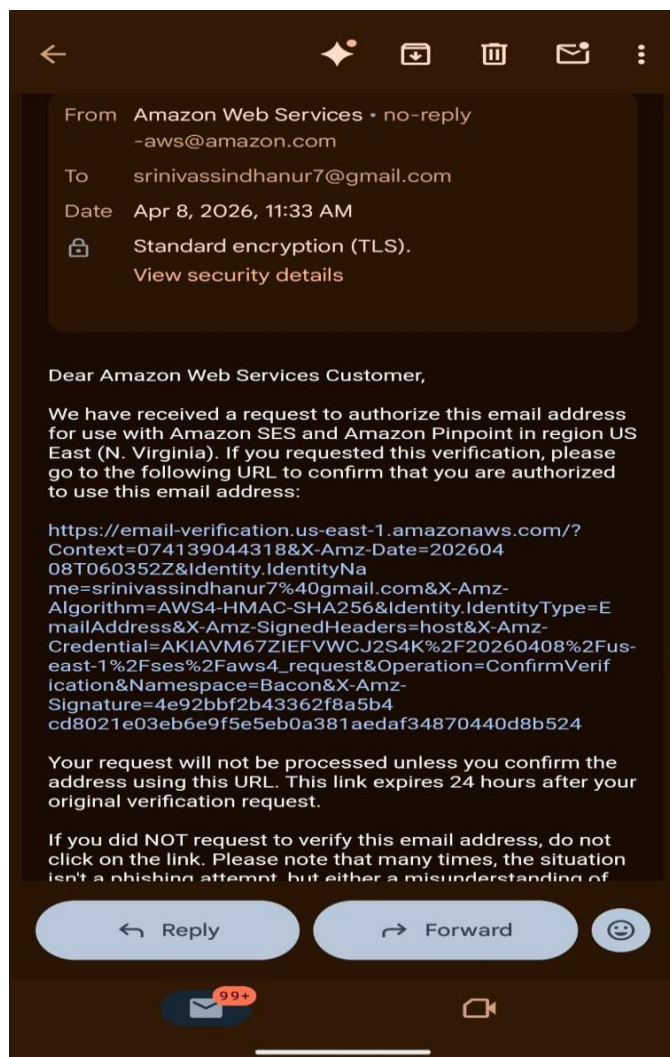
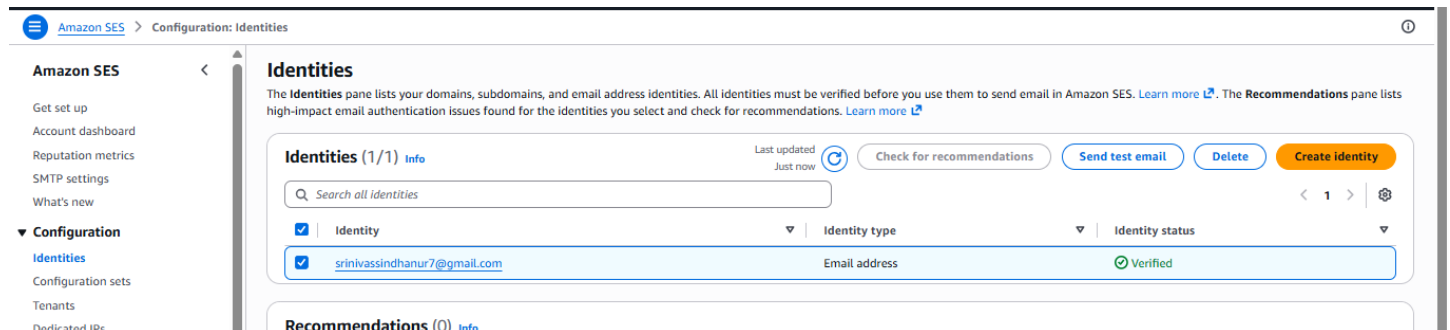
[View in AWS Cost Explorer](#)

Alerts

- ▶ Actual cost > 85% | No actions
- ▶ Forecasted cost > 100% | No actions
- ▶ Actual cost > 100% | No actions

STEP 5: Setup Amazon SES (Simple Email Service)

- Open Amazon SES
 - Region: Mumbai (ap-south-1)
 - Verify Email Address: Go to Verified identities
 - Click Create identity
 - Choose Email address: Enter your email
 - Verify from inbox



STEP 6: Add Permission to Lambda Role

- Open IAM → Roles
- Select Lambda role Attach policy:
 - AmazonSESFullAccess
 - AWSBillingReadOnlyAccess
 - AWSLambdaBasicExecutionRole

The screenshot shows the AWS IAM console for the role 'CostAlertFunction-role-z8sqfng'. The left sidebar shows the navigation menu with 'Identity and Access Management (IAM)' selected. The main content area shows the role's summary, including its ARN and creation date. Below the summary, the 'Permissions' tab is active, displaying a table of attached policies.

Policy name	Type	Attached entities
AmazonSESFullAccess	AWS managed	2
AWSBillingReadOnlyAccess	AWS managed	1
AWSLambdaBasicExecutionRole-a565f4e2-b63b-4f...	Customer managed	1

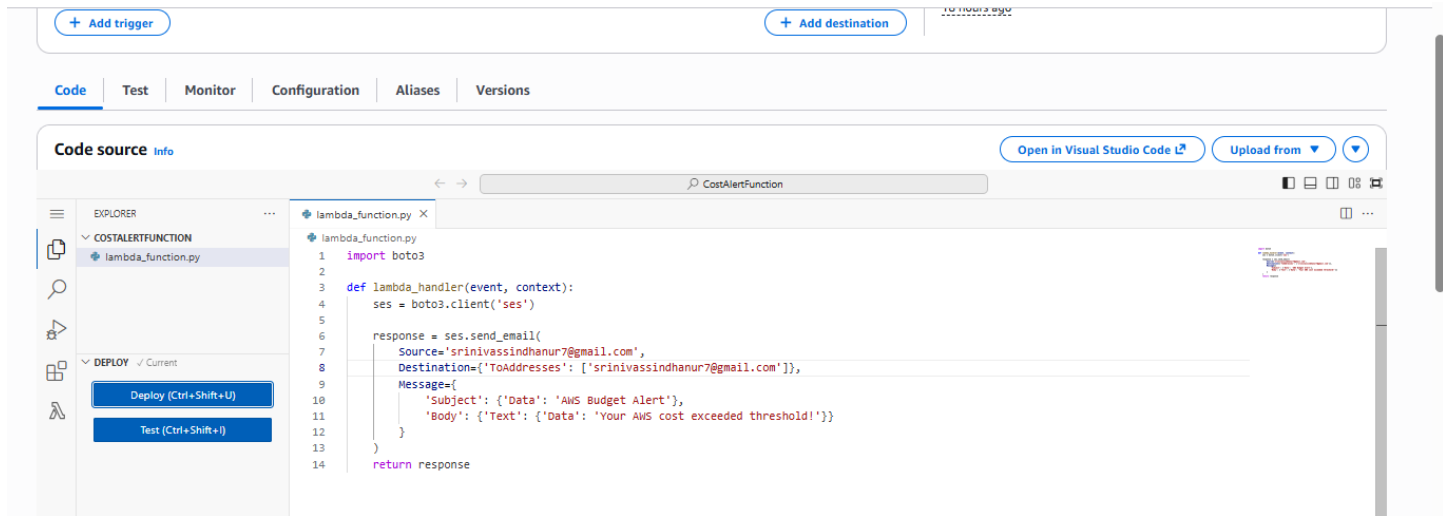
STEP 7: Create Lambda Function

- Open AWS Lambda
- Click Create function Author from scratch:
 - Function name: BudgetAlertHandler
 - Runtime: Python 3.14

Click Create

The screenshot shows the AWS Lambda console for the function 'CostAlertFunction'. The left sidebar shows the navigation menu with 'Lambda' selected. The main content area shows the function's details, including its name, description, package type, runtime, and type.

Function name	Description	Package type	Runtime	Type	Last modified
CostAlertFunction	-	Zip	Python 3.14	Standard	18 hours ago



Troubleshooting:

Cost explorer will not show data immediately after enabling it.

Solution: wait for up to 24 hours because AWS takes time to generate and display billing data. If its first time.

Conclusion:

The AWS Cost Monitoring & Budget Alerts project helps monitor cloud spending and prevent unexpected costs by setting budget limits. It automatically sends notifications when the spending reaches predefined thresholds. This project improves cost visibility and enables better financial control of AWS resources. Overall, it demonstrates how AWS services can be used to build an efficient automated cost management system.